Swaham Pitambar Patra

Task 1/2:

```python
# Q1 and Q2
class BankAccount:
    def __init__(self, balance=0):
        self.__balance = balance  # private attribute

    def deposit(self, amount):
        self.__balance += amount
        print(f"Deposited {amount}. New balance: {self.__balance}")

    # Q1: Add withdraw method
    def withdraw(self, amount):
        if amount <= self.__balance:
            self.__balance -= amount
            print(f"Withdrew {amount}. New balance: {self.__balance}")
        else:
            print("Insufficient funds!")

# Object Initialization and function calling.
account1 = BankAccount(1000)

account1.deposit(500)

account1.withdraw(300)  # Withdral Successful
account1.withdraw(1500)  # Should show insufficient funds

# Q2: Try to access __balance from outside
print(account1.__balance)
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 08-07\task1,2.py"
Deposited 500. New balance: 1500
Withdrew 300. New balance: 1200
Insufficient funds!
Traceback (most recent call last):
  File "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 08-07\task1,2.py", line 27, in <module>
    print(account1.__balance)
          ^^^^^^^^^^^^^^^^^^
AttributeError: 'BankAccount' object has no attribute '__balance'
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Task 3:

```python
# Q3
class Person:
    def greet(self):
        print("Hello!")

class Teacher(Person):
    def teach(self):
        print("Teaching...")

# Object Initialization and function calling.
t = Teacher()
t.greet()    # Inherited
t.teach()    # Own method
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham
Hello!
Teaching...
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Task 4:

```python
# Q4
class Flyer:
    def fly(self):
        print("Flying high!")

class Swimmer:
    def swim(self):
        print("Swimming deep!")

class FlyingFish(Flyer, Swimmer):
    pass  # Inherits from both

# Object Initialization and function calling.
f = FlyingFish()
f.fly()
f.swim()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham P
Flying high!
Swimming deep!
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Task 5:

```python
# Q5
class Manager:
    def greet(self):
        print("Hello, I am a Manager.")

class Director(Manager):
    def greet(self):
        print("Hello, I am a Manager (Director).")  # Overridden

# Object Initialization and function calling.
d = Director()
d.greet()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS
Hello, I am a Manager (Director).
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Task 6:

```python
# Q6
class Principal:
    def manage_school(self):
        print("Managing the school.")

class Teacher(Principal):
    def teach(self):
        print("Teacher is teaching.")

class Student(Principal):
    def study(self):
        print("Student is studying.")

# Object Initialization and function calling.
t = Teacher()
s = Student()

t.manage_school()  # Inherited
t.teach()
s.manage_school()  # Inherited
s.study()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham
Managing the school.
Teacher is teaching.
Managing the school.
Student is studying.
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```

Task 7:

```python
# Q7
class Device:
    def power_on(self):
        print("Device powered on.")

class Phone(Device):
    def call(self):
        print("Making a call.")

class Camera(Device):
    def take_photo(self):
        print("Taking a photo.")

class SmartPhone(Phone, Camera):
    pass  # Multiple inheritance

# Object Initialization and function calling.
sp = SmartPhone()

sp.power_on()
sp.call()
sp.take_photo()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaha
Device powered on.
Making a call.
Taking a photo.
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```